

RNA-seq Unimodal Molecular Subtype Classification of TCGA-BRCA

1 Problem Definition

We evaluate a unimodal RNA-seq branch for **PAM50 molecular subtype classification** in the TCGA-BRCA cohort. This branch is intended as a molecular baseline and future fusion component for the existing histology branch.

The target consists of four PAM50 molecular subtypes:

Table 1: PAM50 classes used for the RNA-seq experiment.

Class	Label	Description
0	Luminal A	Hormone receptor-positive, low proliferation
1	Luminal B	Hormone receptor-positive, higher proliferation
2	Basal-like	Triple-negative, aggressive
3	HER2-enriched	HER2 overexpression

Samples labelled as Normal-like were excluded to match the current four-class WSI setup. The RNA-seq table contains 1,045 primary tumour samples. Each fold uses 837 training samples, 104 validation samples, and 104 test samples.

2 Pipeline

The classification pipeline follows a standard tabular molecular-learning setup.

2.1 Stage 1: RNA-seq Preprocessing

RNA-seq data were downloaded, filtered to the target PAM50 classes, converted into CLAM-compatible sample tables, and aligned with the patient-level fold splits. For the selected configuration, gene features are selected within each training fold by variance and standardised using training-fold statistics only.

2.2 Stage 2: RNA-level Classification – MLP

Sample-level predictions are made using a fully connected multilayer perceptron. The best configuration selected by the sweep uses:

Table 2: Best RNA configuration selected by Bayesian sweep on fold 0 validation AUC $\uparrow$.

Parameter	Value
Architecture	MLP
Hidden dimensions	[1024, 512]
Input genes	10000 top-variance genes
Dropout	0.4
Optimizer	AdamW
Learning rate	0.000197465
Weight decay	0.000700548
Label smoothing	0.05
Weighted loss	Enabled
Weighted sampling	Enabled
Batch size	64
Max epochs	80
Early stopping patience	16

3 Experimental Setup

3.1 Hyperparameter Optimisation

Hyperparameters were tuned via Weights & Biases Bayesian sweeps, maximising validation AUC $\uparrow$ on fold 0. The search space included learning rate, weight decay, dropout, hidden dimensions, number of selected genes, label smoothing, batch size, weighted loss, weighted sampling, and early stopping patience.

3.2 Evaluation Protocol

The best configuration was evaluated using stratified 10-fold cross-validation with patient-level splitting. Each patient appears in exactly one test fold. Aggregating predictions across the ten folds therefore yields held-out predictions for the full RNA cohort without leakage across train and test partitions.

4 Results

4.1 Per-Fold Validation and Test Metrics

Each fold used 837 training, 104 validation, and 104 test samples with 10,000 selected genes. Table 3 shows the per-fold validation and test metrics.

Table 3: Per-fold RNA-seq validation and test results.

Fold	Val AUC $\uparrow$	Val Acc. $\uparrow$	Test AUC $\uparrow$	Test Acc. $\uparrow$
0	0.9982	0.9519	0.9711	0.9038
1	0.9911	0.9615	0.9873	0.9135
2	0.9922	0.9231	0.9828	0.8942
3	0.9877	0.9038	0.9924	0.9135
4	0.9888	0.8942	0.9839	0.8750
5	0.9784	0.8750	0.9751	0.8846
6	0.9918	0.8942	0.9869	0.8942
7	0.9947	0.9038	0.9858	0.9135
8	0.9897	0.9231	0.9757	0.8654
9	0.9841	0.8750	0.9767	0.8558
Mean $\pm$ SD	0.9897 ± 0.0055	0.9106 ± 0.0294	0.9818 ± 0.0068	0.8913 ± 0.0208

4.2 Aggregated Predictions

Since each RNA sample appears in exactly one test fold, we aggregate all test predictions across the 10 folds to obtain held-out predictions for the full cohort. The aggregate held-out test performance is:

- Macro AUC $\uparrow$: 0.9821
- Overall accuracy $\uparrow$: 0.8913
- Weighted F1-score $\uparrow$: 0.8930

4.3 Per-Fold Test AUC $\uparrow$

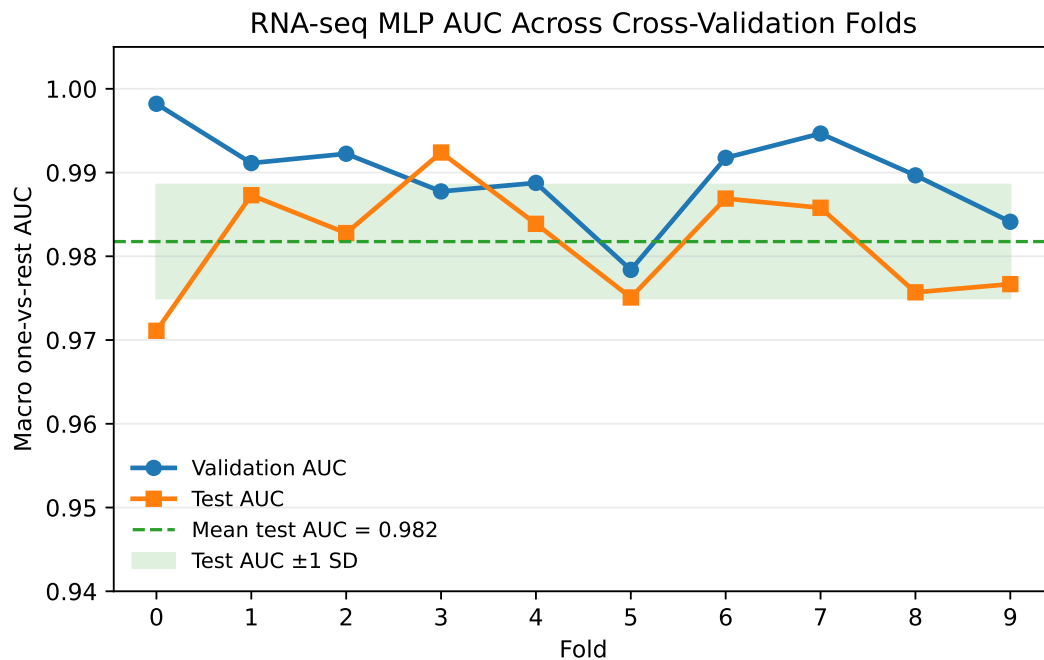


Figure 1: Validation and test AUC $\uparrow$ across folds. The dashed line denotes mean test AUC $\uparrow$ and the shaded region denotes one standard deviation.

4.4 Confusion Matrix

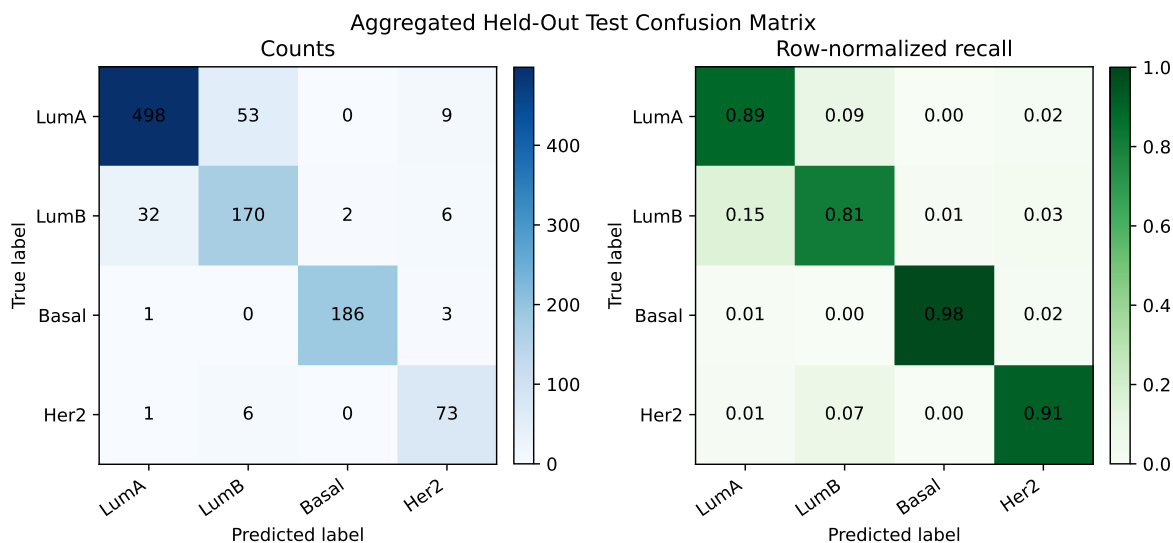


Figure 2: Confusion matrix for aggregated 10-fold held-out test predictions. Left: raw counts. Right: row-normalised recall $\uparrow$.

4.5 Classification Report

Table 4: Per-class test metrics from aggregated held-out predictions.

Class	AUC $\uparrow$	Precision $\uparrow$	Recall $\uparrow$	F1 $\uparrow$	Support
LumA	0.9782	0.936	0.889	0.912	560
LumB	0.9621	0.742	0.810	0.774	210
Basal	0.9991	0.989	0.979	0.984	190
Her2	0.9889	0.802	0.912	0.854	80

4.6 ROC Curves

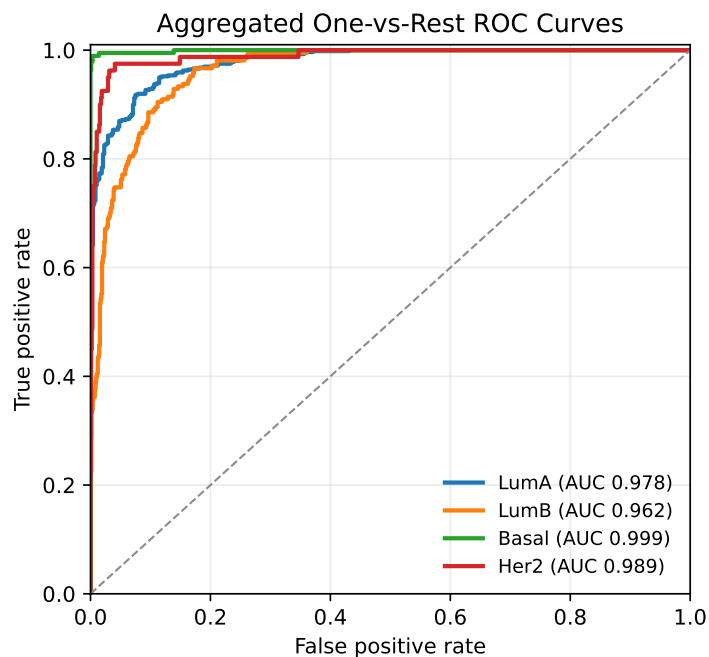


Figure 3: One-vs-rest ROC curves computed from aggregated held-out test predictions.

4.7 Per-Class AUC $\uparrow$ Across Folds

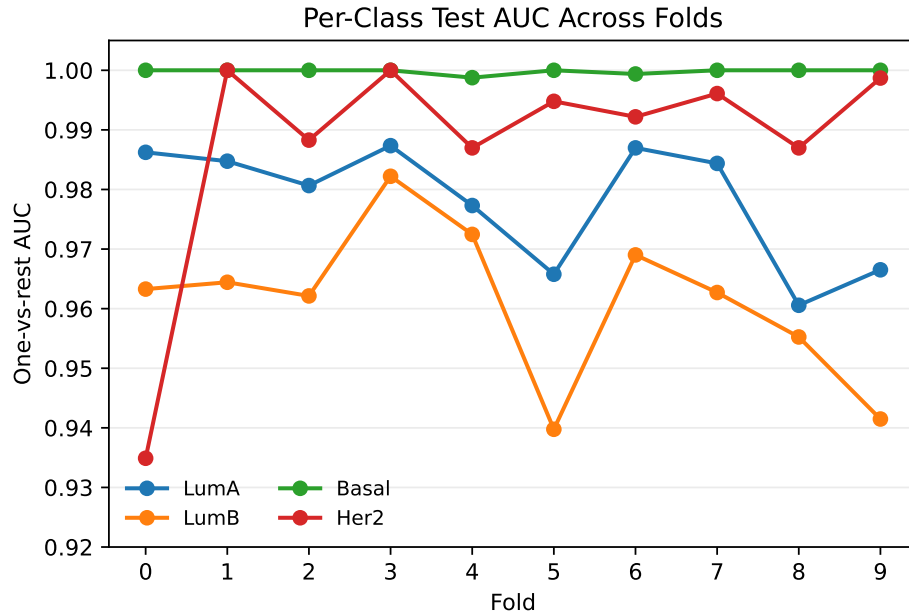


Figure 4: Per-class test AUC $\uparrow$ across the 10 folds.

5 Comparison to the WSI-only Branch

The RNA-only branch is compared with the existing WSI-only CLAM-MB + UNI2-h branch. The WSI branch remains the primary thesis focus because it predicts molecular subtype from histology. RNA-seq is used here as a strong molecular reference and future fusion modality. For the visual comparison, we use macro AUC, balanced accuracy, and macro F1 because the task is four-class and class-imbalanced. WSI balanced accuracy and macro F1 are computed from the aggregated per-class recall and F1 values.

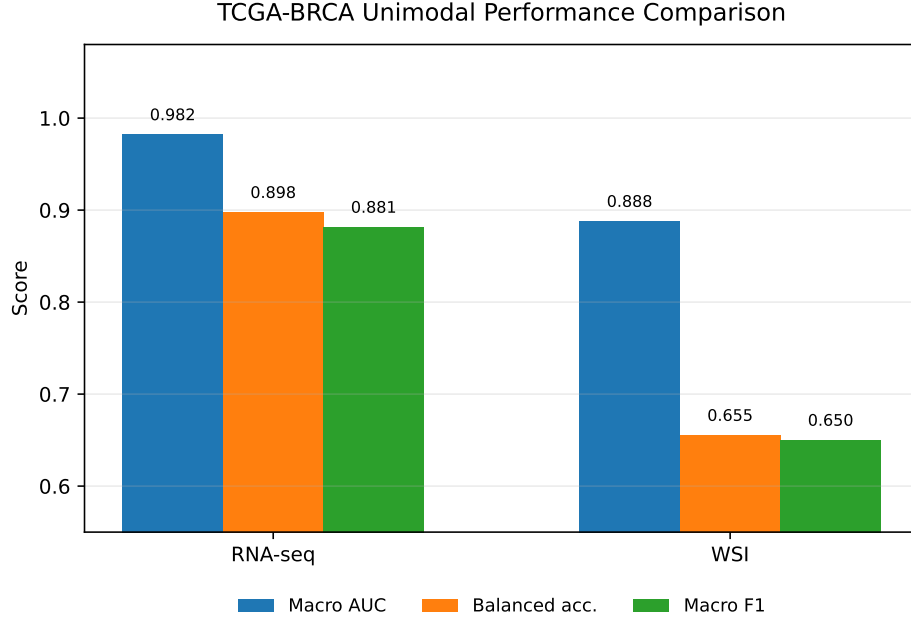


Figure 5: Comparison of RNA-seq and WSI unimodal branches using class-sensitive metrics. Higher values are better.

6 Comparison to SOTA

Hezil et al. proposed *Selective multimodal deep learning for reliable breast cancer subtype classification from histopathology and genomic data* [1], a TCGA-BRCA PAM50 pipeline combining CTransPath WSI features, RNA-seq features, multimodal fusion, and confidence-aware routing. Tables 5 and 6 separate the single-modality and multimodal comparisons.

Table 5: Single-modality comparison on TCGA-BRCA PAM50 subtype classification. Literature metrics are from Hezil et al. [2]. Best values are bolded within comparable modality settings.

Method	Setting	Cohort	Accuracy $\uparrow$	Macro F1 $\uparrow$	Weighted F1 $\uparrow$
<i>Reference methods</i>					
Hezil et al. [2]	RNA-seq only	1022 patients	0.9100	0.90	0.91
Hezil et al. [2]	WSI only, CTransPath	1022 patients	0.6200	0.57	0.64
<i>Our methods</i>					
RNA-seq	RNA-seq MLP	1045 samples	0.8913 ± 0.0208	0.8815 ± 0.0292	0.8929 ± 0.0198
Vision Branch	UNI2-h + CLAM-MB	986 slides	0.7181	0.6500	0.7179

Table 6: Multimodal comparison on TCGA-BRCA PAM50 subtype classification.

Method	Setting	Cohort	Accuracy $\uparrow$	Macro F1 $\uparrow$	Weighted F1 $\uparrow$
Hezil et al. [2]	Multimodal concatenation	1022 patients	0.9350	0.92	0.93
Hezil et al. [2]	Multimodal gated fusion	1022 patients	0.9420	0.93	0.94
Hezil et al. [2]	Multimodal cross-attention	1022 patients	0.9480	0.93	0.95
Hezil et al. [2]	Routing-based fusion	1022 patients	0.9505	0.93	0.95
Hezil et al. [1]	Routing-based fusion	924 patients	0.9493	—	—

RNA-seq is the stronger single-modality baseline for PAM50, while multimodal fusion gives the highest reported accuracies in Hezil et al. These values are not a direct ranking because cohort size, splits, feature extractors, and reporting protocols differ.

This result is consistent with prior work showing that RNA and other omics modalities contain strong subtype signal. Lin et al. [4] proposed DeepMO for breast cancer subtype classification from mRNA, methylation, and copy-number alteration data, reporting that mRNA and methylation are highly informative for subtype classification. Ektefaie et al. [3] studied histology-only and multiomics-histopathology breast cancer classification and reported four-class PAM50 prediction from H&E images, supporting the histology-first framing. Recent multimodal studies further support attention- or routing-based fusion of image and molecular or clinical modalities [5], although the precise datasets, subtype groupings, and validation protocols differ.

7 Summary and Interpretation

The RNA-seq branch achieved strong 10-fold performance on TCGA-BRCA four-class PAM50 classification:

- Validation macro AUC $\uparrow$: 0.9897 ± 0.0055
- Validation accuracy $\uparrow$: 0.9106 ± 0.0294
- Test macro AUC $\uparrow$: 0.9818 ± 0.0068
- Test accuracy $\uparrow$: 0.8913 ± 0.0208
- Test balanced accuracy $\uparrow$: 0.8976 ± 0.0284

This performance is consistent with the fact that PAM50 is defined from gene expression. In our approach, the RNA-only model serves as a molecular reference branch and as the basis for later fusion with WSI features. The histology branch addresses the main image-based task: predicting molecular subtype from H&E whole-slide images, with the RNA branch providing a complementary modality for multimodal learning.

References

1. Hezil, N. et al. Selective multimodal deep learning for reliable breast cancer subtype classification from histopathology and genomic data. *Medical Engineering & Physics* (2025). [PubMed record](#).

2. Hezil, N. et al. Selective multimodal breast cancer subtype classification implementation and reported metrics. [Code repository](#).
3. Ektefaie, Y. et al. Integrative multiomics-histopathology analysis for breast cancer classification. *npj Breast Cancer* 7, 147 (2021). [Article](#).
4. Lin, Y. et al. Classifying Breast Cancer Subtypes Using Deep Neural Networks Based on Multi-Omics Data. *Genes* 11(8), 888 (2020). [Article](#).
5. Ben Rabah, C. et al. A Multimodal Deep Learning Model for the Classification of Breast Cancer Subtypes. *Diagnostics* 15(8), 995 (2025). [Article](#).